

Driver Monitoring System Using Transfer Learning for Edge Deployment

Description

This mini-project develops a compact driver monitoring system using transfer learning, targeting real-time deployment on edge devices like mobile phones and Raspberry Pi. The goal is to achieve >90% accuracy across five driver states (sleepy, reckless, aware, drinking, yawning) while keeping the model under 4 MB with inference times suitable for embedded hardware.

Methodology

- **Transfer Learning Approach:** Comparative analysis of state-of-the-art architectures (VGG16, EfficientNetB1, MobileNetV3, NASNet) for a 6-class classification task.
- **Edge Optimization Focus:** Model compression via TensorFlow Lite, input resolution tuning, and layer freezing/unfreezing strategies to balance accuracy and latency.
- **Experimental Rigor:** All experiments are conducted in reproducible notebooks with fixed random seeds, loss/accuracy charts, and ablation studies to document design decisions.

Key Findings (Current work)

- VGG16 achieved 96% accuracy at 224×224 resolution but is unsuitable for edge deployment (~400 ms inference even with XNNPACK CPU acceleration).
- MobileNetV3 collapsed to 30–31% accuracy at 64×64 due to spatial resolution loss (feature maps reduced to 2×2 before classification).
- **Critical Insight:** MobileNetV3 requires minimum 96×96 input; planned testing at 128×128 aims to preserve facial details while reducing model size.

Target Output

- Final model in **.tflite format** under **4 MB**, optimized for edge inference with >90% accuracy.
- Benchmark results comparing latency (ms) and accuracy across tested architectures.

References

- Al-Gburi, S. H., Al-Sammak, K. A., Marghescu, I., Oprea, C. C., Drăgulescu, A.-M. C., Alduais, N. A. M., Alheeti, K. M. A., & Al-Sammak, N. A. H. (2025). EffRes-DrowsyNet: A Novel Hybrid Deep Learning Model Combining EfficientNetB0 and ResNet50 for Driver Drowsiness Detection. *Sensors*, 25(12), 3711. <https://doi.org/10.3390/s25123711>
- Ademola, O. A., Leier, M., & Petlenkov, E. (2021). Evaluation of Deep Neural Network Compression Methods for Edge Devices Using Weighted Score-Based Ranking Scheme. *Sensors*, 21(22), 7529. <https://doi.org/10.3390/s21227529>
- Hohman, F., Kery, M. B., Ren, D., & Moritz, D. (2024). Model Compression in Practice: Lessons Learned from Practitioners Creating On-device Machine Learning Experiences. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Article 645). Association for Computing Machinery. <https://doi.org/10.1145/3613904.3642109>
- Khan, T., Choi, G., & Lee, S. (2023). EFFNet-CA: An Efficient Driver Distraction Detection Based on Multiscale Features Extractions and Channel Attention Mechanism. *Sensors*, 23(8), 3835. <https://doi.org/10.3390/s23083835>

Next Steps

1. Train and evaluate MobileNetV3-Small at 128×128 input.
2. Quantize and convert best-performing model to TFLite.
3. Validate on edge hardware (Raspberry Pi 4/Android phone).
4. Document trade-offs and final performance metrics.